Uday Gandhi

Shegaon, Maharashtra, India | +91-7020734678 | udaygandhi2004@gmail.com
LinkedIn | GitHub | Portfolio

Professional Summary

Computer Science graduate with Honors in Data Science (8.81/10.0 CGPA) who builds software products and explores AI/ML through hands-on projects. Experience spans full-stack development, REST APIs, audio signal processing, deep learning with PyTorch, and data-intensive pipelines. Skilled at taking ideas from problem definition through implementation and evaluation, with a long-term focus on building practical AI-powered products.

Education

Shri Ramdeobaba College of Engineering and Management, Nagpur **2022–2026**
B.Tech in Computer Science (Honors in Data Science) CGPA: 8.81 / 10.0

Shri Gajanan Maharaj English School, Shegaon **2021–2022**
Class XII, State Board Percentage: 82.3%

Shri Gajanan Maharaj English School, Shegaon **2019–2020**
Class X, State Board Percentage: 94.4%

Technical Skills

Programming & Query Languages: SQL, Python, Java, C, JavaScript
Data Science & Statistics: Regression, Classification, Clustering, Hypothesis Testing, Data Preprocessing, Model Optimization
Data Engineering & Databases: Large-Scale Data Processing, Data Pipelines, MongoDB, MySQL
Machine Learning: PyTorch, scikit-learn, NumPy, OpenCV, Deep Learning, Image Segmentation
Cloud & Development: Amazon AWS, Flask, Django, Node.js, Express, React
Developer Tools: Git, GitHub, Blender

Experience

Cognizant **Jan 2026 – May 2026**
MERN Stack Developer Intern / Technology Training *Chennai, India*

- Architected a secure full-stack Library Management System using MERN, structuring application data and workflows around distinct administrator and patron requirements.
- Engineered scalable RESTful APIs with Node.js and Express to process book inventory, issue/return records, and real-time availability data.
- Designed data-driven interfaces with React and optimized application state management for efficient interaction with backend data.

Selected Projects

Stemify *AI-Based Audio Source Separator*

- Engineered a U-Net deep learning model using PyTorch for audio source separation, isolating vocals and instruments from mixed tracks.
- Built an end-to-end analytical pipeline covering signal preprocessing, model inference, and automated post-processing.
- Deployed a Flask REST API for concurrent processing requests, enabling automated on-demand audio analysis.
- Integrated STFT-based tempo detection and automated sheet-music transcription, exporting analytical results to PDF/XML.

NotShazam *Music Fingerprinting & Identification System*

- Architected a scalable data-processing pipeline over a 350+ song library, generating and indexing 11M+ unique spectrogram hashes in MongoDB.
- Transformed raw audio signals into structured fingerprint representations, enabling efficient large-scale matching and retrieval.
- Optimized the matching algorithm to achieve sub-200ms latency for real-time audio identification from sources including YouTube streams.
- Developed robust hashing techniques to maintain accurate matching performance under noise and signal distortion.

OvaCare *Ultrasound Image Analysis for PCOS Detection*

- Developed an automated image-analysis pipeline using PyTorch, OpenCV, and SAM-based segmentation.
- Implemented data preprocessing, image upscaling, segmentation-mask filtering, and output refinement to improve analytical consistency.
- Automated detection, counting, and measurement of follicles and cysts, improving the reproducibility of the analytical workflow.

Certifications & Achievements

- Solved 400+ DSA problems on LeetCode, strengthening algorithmic problem-solving and analytical reasoning skills.
- DeepLearning.AI – Generative Adversarial Networks (GANs) Specialization
- Cisco – Python Essentials
- Google – Project Management